TASK 5 - STUDENT PERFORMANCE PREDICTION

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Course: Artificial Intelligence

Week: 3

Mini Project: Student Performance Prediction

Objective

The objective of this mini-project is to build a simple machine learning classification system that predicts whether a student is likely to PASS or FAIL based on academic and attendance-related features.

Dataset and Features

A dataset containing 50 student records was created using Python and Pandas. The selected features were Study Hours, Attendance, Previous Marks, Assignment Score, and Project Score. The target variable was Result, containing two classes: PASS and FAIL.

Data Preparation

The dataset was checked for missing values and data types. No missing values were found. The target variable was encoded using LabelEncoder, with FAIL represented as 0 and PASS represented as 1. The dataset was divided into training and testing data using an 80:20 ratio. StandardScaler was applied to standardize the numerical features.

Model and Results

A Logistic Regression classification model was selected because the project involves predicting one of two categories. The model was trained using the training data and evaluated using the testing data. The model achieved an accuracy of **100%** on the testing dataset. A classification report was also generated to evaluate precision, recall, and F1-score.

Testing New Students

The model was additionally tested using three new student inputs. These students were provided with different study hours, attendance percentages, previous marks, assignment scores, and project scores, and the trained model generated PASS or FAIL predictions.

Limitations and Future Improvements

One limitation of this project is the relatively small dataset. The 100% test accuracy should therefore not be interpreted as proof that the model will always perform perfectly on real-world students. Future improvements could include collecting a larger and more diverse dataset, adding additional student-related features, comparing different machine learning algorithms, and evaluating the model using cross-validation.

Project Conclusion

The project demonstrates the fundamental machine learning workflow of data preparation, preprocessing, model training, prediction, and evaluation. It provides a foundation for developing more robust student-performance prediction systems with larger real-world datasets.